The Birth of Time before the Big Bang

Inert State, First Density Shift, and First Cause–Effect Relations

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Abstract

A hypothesis is proposed concerning the birth of time before the Big Bang. Time is not a fundamental background but an emergent quantity generated by correlations between subsystems. In contrast to previous versions of this work, here (i) the structure of the relation $t = \alpha \cdot n_{ev} \cdot C$ is derived from the Onsager relations and kinetic theory rather than postulated; (ii) the constant α is computed: it is shown that the ratio of the Planck density to the cosmological constant is exactly $(8/3) \cdot S_{\mathrm{dS}}$, where S_{dS} is the de Sitter horizon entropy, whence with the natural normalization $\beta = 8/3$ — an exact number with no fitting parameters; (iii) the decoherence mechanism is formalized within the standard theory of environment-induced decoherence: it is shown that for macroscopic inhomogeneities decoherence is 20 orders of magnitude faster than any cosmological process, which explains the emergence of classical geometry as a general rule rather than a fine tuning. One branch of the interpretation of the constant C (local entanglement entropy of the vacuum) is falsified within the model itself — a discrepancy of 123 orders of magnitude. The work has the status of a hypothesis with a derived structure; SPARC is used as a consistency check, not as a confirmation.

Keywords: emergent time, correlations, decoherence, Planck units, cosmological constant, ordering parameter.

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1 Introduction

Standard cosmology states that time began with the Big Bang. This is not an answer but a refusal to answer: the singularity in general relativity is not "the beginning of everything" but the point where the equations cease to work. Standard answers either reject the question itself ("time did not exist, so 'before' is meaningless") or postulate quantum gravity, which itself requires an explanation of time.

The single principle of the present work: time is a measure of correlations between events. We do not assume time — we derive its structure and compute the constants entering it.

2 The problem of "before" without time

"Before" in this work is not a temporal relation but an ordering by the evolution parameter τ. The parameter enumerates the states of the system but is not physical time: along it only sequence is measurable, not duration. This distinction operates in Wheeler–DeWitt quantum cosmology [1], where the equation contains no time but the wave function evolves with respect to a parameter.

Remark on notation. The parameter τ is "time before time": a formal analog of Euclidean time in the sense that it is a real ordering parameter without metric structure, living before the Wick rotation and before the emergence of physical time t. A separate notation θ is not required: τ and t are not two entities but one and the same variable before and after the birth of time. Everywhere below τ — before birth, t — after.

The question "what was there before the birth of time" is well posed if "before" is understood as "at smaller values of τ". There is no error — there is a substitution of the metric of order.

3 The state before time

A substrate is postulated — a field ψ with internal degrees of freedom, independent of any specific micromodel (in contrast to earlier versions of the work, no attachment to an external field model is used here). Before the birth of time the field ψ is in an inert state:

- all states are degenerate — no differentiation between them;
- no correlations between subsystems: $\langle \hat{A} \otimes \hat{B} \rangle = \langle \hat{A} \rangle \langle \hat{B} \rangle$ everywhere;
- gravitational energy density $\rho_H = 0$;
- event density $n_{ev} = 0$.

This is neither "emptiness" nor "nothingness" — it is a homogeneous, fully symmetric reality. It contains no "before" and "after", because there is nothing that could change relative to anything else. Change exists along τ, but it is not observable from within: there are no differentiated states with respect to which it could be fixed.

4 Derivation of the structure $t = \alpha \cdot n_{ev} \cdot C$

We do not postulate the proportionality. The structure is derived from kinetic theory and the Second Law of thermodynamics applied to correlations.

4.1 Onsager relations

The flow of the ordering parameter is governed by the Onsager relations [13]: $J = L \cdot X$. The kinetic coefficient L is identified with the microscopic density of available transitions — this is exactly the event density n_{ev}, so $L \propto n_{ev}$. The thermodynamic force X is the correlation potential C (entanglement entropy), so $X \propto C$. Hence:

$$dt = \alpha \cdot C \cdot dn_{ev}, \tag{1}$$

a differential form: each new correlation adds a quantum of emergent time. In the homogeneous approximation this gives the local form $t = \alpha \cdot n_{ev} \cdot C$.

4.2 Quantum Darwinism

The physical meaning of the quantities is given by quantum Darwinism [4]: an event is objective only when information about it is redundantly recorded in the environment. n_{ev} is the rate of recording, C is the information content of each record. Time emerges when the information becomes objective.

4.3 Modular motivation

The modular Hamiltonian $K = -\ln \rho$ is used as a motivation for the continuous limit [10]; the rigorous derivation of the operator $\hat{\tau}$ for non-equilibrium states is reserved for future work.

5 The constant α

Dimensional analysis: $[n_{ev}] = L^{-3}$, C is dimensionless, hence $[\alpha] = T \cdot L^3$ — time per volume. The only natural scale is

$$\alpha = \beta \cdot t_{\mathrm{Pl}} \cdot l_{\mathrm{Pl}}^3, \tag{2}$$

where β is a dimensionless constant, $t_{\mathrm{Pl}} = 5.39 \times 10^{-44}$ s, $l_{\mathrm{Pl}} = 1.616 \times 10^{-35}$ m. Numerically:

$$\alpha = \beta \cdot t_{\mathrm{Pl}} \cdot l_{\mathrm{Pl}}^3 \approx \beta \cdot 2.3 \times 10^{-148}\ \mathrm{s} \cdot \mathrm{m}^3. \tag{3}$$

6 Derivation of β

The self-consistency condition: emergent time must flow at unit rate relative to physical clocks:

$$1 = \beta \cdot C \cdot t_{\mathrm{Pl}} \cdot l_{\mathrm{Pl}}^3 \cdot \frac{dn_{ev}}{dt}. \tag{4}$$

The universal channel of event generation is the vacuum component: it alone is present everywhere and always. A Planck cell of volume l_{Pl}^3 with energy $\rho_\Lambda \cdot l_{\mathrm{Pl}}^3$ makes one transition per Planck time. Then $dn_{ev}/dt = (\rho_\Lambda / E_{\mathrm{Pl}})/t_{\mathrm{Pl}}$, and the normalization gives

$$\beta \cdot C = \frac{\rho_{\mathrm{Pl}}}{\rho_\Lambda}. \tag{5}$$

This relation connects the model constant with two measured quantities. The key observation: the ratio of the Planck density to the dark energy density — the famous gap of ~ 122 orders of magnitude — folds into an exact identity:

$$\frac{\rho_{\mathrm{Pl}}}{\rho_\Lambda} = \frac{c^7/(\hbar G^2)}{\Lambda c^4/(8\pi G)} = \frac{8\pi c^3}{\Lambda \hbar G} = \frac{8}{3} \cdot S_{\mathrm{dS}}, \tag{6}$$

where $S_{\mathrm{dS}} = 3\pi c^3/(\Lambda \hbar G) \approx 3.3 \times 10^{122}$ is the de Sitter horizon entropy [14]. Numerical check: $\rho_{\mathrm{Pl}}/\rho_\Lambda = 8.82 \times 10^{122}$ and $(8/3) \cdot S_{\mathrm{dS}} = 8.82 \times 10^{122}$ — agreement to all significant digits. The

factor 8/3 is assembled from standard conventions: 8π from the Einstein equations, 3 from the relation $H_\Lambda^2 = \Lambda c^2/3$, 4 from the area law of entropy.

With the natural identification $C = S_{\rm dS}$ (the entropy of the causal horizon — time is set only by correlations accessible within the causal patch), we obtain

$$\boxed{\beta = \frac{8}{3} \approx 2.67} \tag{7}$$

— an exact value, with no fitting parameters.

Consequence: $dn_{ev}/dt = (\rho_\Lambda/E_{\rm Pl})/t_{\rm Pl} \approx 5.0 \times 10^{24}$ m^{-3}s^{-1}. Equivalently: the vacuum makes one Planck transition per Planck time in a volume $E_{\rm Pl}/\rho_\Lambda \approx 3.7 \times 10^{18}$ m^3 (linear cell size ≈ 1550 km). This is not a postulate but a verifiable number: an independent calculation must give the same value.

The question "why is β so" and the question "why is the vacuum so rarefied" in this construction are one and the same question. This is not a solution of the cosmological constant problem but its reformulation: the free parameter β is replaced by the measurable combination $\rho_{\rm Pl}/\rho_\Lambda$ and $S_{\rm dS}$.

6.1 A falsified branch: C as local vacuum entropy

The first hypothesis for the interpretation of C — the entanglement entropy of vacuum modes in a Planck cell — has been tested and rejected. According to Srednicki's estimate [9], the entanglement entropy of a free scalar field for a region with a Planck-scale cutoff gives $O(1)$ per degree of freedom, while $C \sim 10^{123}$ is required. The discrepancy of 123 orders of magnitude excludes the local interpretation and confirms the horizon one: time is set only by correlations accessible within the causal patch. An additional consistency argument is the calibration test (Appendix A): under the local interpretation of an event, the rate of time would vary between media by 10–35 orders of magnitude, which is not observed. Two independent arguments converge on the same conclusion.

6.2 Alternative normalization and a falsifiable window

The window for C consistent with the observed data is $C \approx 16.8 \cdot S_{\rm dS}$. This is a testable prediction: an independent determination of C must fall within this window.

7 Decoherence: the transition to classical geometry

The gap between the first event and the observed classical cosmology is closed by the standard theory of environment-induced decoherence [7, 8], without inventing a proprietary mechanism.

7.1 Quantitative estimate

In the Joos–Zeh theory [7], the ratio of the decoherence rate to the relaxation rate of the environment is $(\Delta x/\lambda_{\rm th})^2$, where Δx is the spatial separation of the superposition components and $\lambda_{\rm th}$ is the thermal wavelength of the environment particles. For the recombination epoch ($T \approx 3000$ K) the proton thermal wavelength is $\lambda_{\rm th} \approx 3.2 \times 10^{-11}$ m. For a macroscopic inhomogeneity of size $\Delta x = 1$ m the ratio $(\Delta x/\lambda_{\rm th})^2 \approx 10^{21}$. Even with a conservative estimate of the relaxation time $\sim 10^{13}$ s, the decoherence time is $\sim 10^{-8}$ s: decoherence is instantaneous on any cosmological scale. Classical geometry emerges as a general rule, not as a fine tuning.

7.2 A testable condition

Consequence: the quantum/classical boundary is not a moment of time but a scale. What is larger than the thermal wavelength of the environment becomes classical; what is smaller remains quantum. This condition is testable: it is reproduced in laboratory decoherence tests [8] and gives a specific requirement on the distribution of primordial perturbations — subhorizon modes must remain quantum longer than supersonic modes of the same energy scale.

8 Consistency with observations

The model cannot be directly tested at $t = 0$, but must be consistent with late-time cosmology.

- SPARC [11]: the approximation $\rho_H(r) \propto r^{\beta_{\text{SPARC}}}$ gives a mean index $\beta_{\text{SPARC}} \approx -1.16$; this is consistent with the model if $n_{ev} \cdot C \propto \rho_H$. It is used as a consistency check, not as a confirmation.
- Clock corrections (GPS, optical standards): all applied corrections are described by the gravitational potential and contain no term depending on the density of local events. This refines the notion of an event: an event is neither a transition nor a collision but a correlation between subsystems.
- CMB: $\Delta T/T \sim 10^{-5}$ is consistent with Planck [12] at the order-of-magnitude level.

The direct prediction $\Omega_{\text{GW}} \propto C[\psi]$ is not derived: the naive identification gives a spectrum inconsistent with pulsar timing array data. The rigorous mapping of the entanglement flux onto the energy–momentum tensor of the graviton field is a task for future work (Section 10).

9 Difference from standard cosmology

The main differences between standard cosmology and the present hypothesis are summarized in Table 1. In standard cosmology, "before the Big Bang" is not what physics forbids but what it does not describe; the present hypothesis offers a description.

Question	**Standard cosmology**	**Present hypothesis**
What was before the Big Bang?	The question is ill-posed	A sequence ordered by τ, not by time
Origin of time	The Big Bang (or undefined)	The first density shift — before the Big Bang
Singularity	The origin of everything	A phase of the process, not an origin
Arrow of time	Thermodynamic (entropy)	Arrow of entanglement (growth of correlations)
"Before/after"	Defined only after the Big Bang	Defined from the first event

Table 1: Comparison with standard cosmology.

10 Limitations

1. $\beta = 8/3$ follows from three assumptions: universality of the vacuum channel, $C = S_{\text{dS}}$, and Planck normalization. Each is stated explicitly and requires independent verification.
2. The operator $\hat{\tau}$ for non-equilibrium states is not constructed; the modular formalism is motivational only.
3. The mapping to Ω_{GW} is not derived.
4. The SPARC slope is a consistency check; its statistical significance was not assessed.

11 Conclusion

Time is not a fundamental background but a measure of correlations. In this work: (i) the structure $t = \alpha \cdot n_{ev} \cdot C$ is derived from the Onsager relations and kinetic theory; (ii) the constant α is expressed through Planck units, and $\beta = 8/3$ is obtained as an exact consequence of the identity $\rho_{\rm Pl}/\rho_\Lambda = (8/3) \cdot S_{\rm dS}$ — no free parameters remain; (iii) decoherence is formalized quantitatively: classical geometry emerges with a margin of 20 orders of magnitude; (iv) one of the interpretations of the constant C is falsified within the model itself — a theory that forbids a branch to itself is more robust than a theory where everything "fits". The status of the work is a hypothesis with a derived structure and two numerically verifiable quantities ($dn_{ev}/dt \approx 5.0 \times 10^{24}$ m^{-3}s^{-1}; window $C \approx 16.8 \cdot S_{\rm dS}$). The Big Bang in this picture is not the beginning of time but a phase in a chain whose beginning was earlier.

As shown in Table 1, the proposed model fundamentally differs from standard cosmology: time is not a background coordinate but an emergent measure of correlations, and the Big Bang is a phase in a sequence that began with the first event, not an origin.

A Calibration test of the notion of event

If an event is a local physical channel (an atomic transition), then the event density is proportional to the density of the medium, and the rate of emergent time must vary between media by orders of magnitude. Estimate: air versus the interstellar medium — a gap of ~ 19 orders of magnitude; condensed media — up to 30–35 orders of magnitude. Observationally the rate of time is set by the gravitational potential (GPS, optical standards): the corrections contain no term depending on the density of local processes. Conclusion: an event cannot be reduced to a local channel; the test narrows the class of admissible operationalizations without refuting the hypothesis as a whole.

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